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Rear-foot, mid-foot and fore-foot motion during the stance phase of gait

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Abstract

This paper proposes a new protocol designed to track a large number of foot segments during the stance phase of gait with the smallest possible number of markers, with particular clinical focus on coronal plane alignment of the rear-foot, transverse and sagittal plane alignment of the metatarsal bones, and changes at the medial longitudinal arch. The shank, calcaneus, mid-foot and metatarsus were assumed to be 3D rigid bodies. The longitudinal axis of the first, second and fifth metatarsal bones and the proximal phalanx of the hallux were also tracked independently. Skin markers were mounted on bony prominences or joint lines, avoiding the course of main tendons. Trajectories of the 14 markers were collected by an eight-camera motion capture system at 100 Hz on a population of 10 young volunteers. Three-dimensional joint rotations and planar angles were calculated according to anatomically based reference frames. The marker set was well visible throughout the stance phase of gait, even in a camera configuration typical of gait analysis of the full body. The time-histories of the joint rotations and planar angles were well repeatable among subjects and consistent with clinical and biomechanical knowledge. Several dynamic measurements were originally taken, such as elevation/drop of the medial longitudinal arch and of three metatarsal bones, rear-foot to fore-foot rotation and transverse plane deformation of the metatarsus. The information obtained from this protocol, consistent with previous clinical knowledge, enhanced our understanding of the dynamics of the human foot during stance.
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1. Introduction

The human shank and foot complex is an intricate, multi-joint mechanism, which is fundamental for the interaction between the lower limb and ground during locomotion [1]. The critical effect of abnormal foot motion on lower limb function has been demonstrated [2–4]. The quantitative assessment of abnormal function and of the effects of treatment requires a more detailed analysis than that offered by standard gait analysis, which considers the foot as a single rigid segment or a vector. Dynamic modelling of the foot also requires multi-segment motion analysis, which takes into account deformity [5,6]. Special techniques based on X-rays [7] and on more modern MRI [8,9] or videofluoroscopy [10] are not applied routinely because of the invasive data acquisition, the restricted field of measurement, and the intense data reduction. Skeletal tracking in vivo [11] is inappropriate in routine clinical assessments. In vitro measurements on cadavers by simulators of locomotion [12] have been criticised for the lack of realistic conditions.

An increased interest in shank and foot multi-segment kinematics analysis in vivo by stereophotogrammetry [13] is documented in the literature. Initially [14–18], a limited number of segments were analysed, and subsequently mid-foot and fore-foot segments were included in the models [19–29], probably because of the availability of more reliable instrumentation. Even a 19-marker, 9-segment, 8-joint model was proposed [26], though marker-to-bone association and validation was limited. All techniques but one [25] utilised stereophotogrammetry. Only a few addressed explicitly adolescents' and children's feet [26,28]. Another recent technique [30] was not based on standard three-dimensional (3D) kinematics, bur rather on isolated planar angles. Despite X-ray-based association between external markers and underlying bones [19,31] or

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the inclusion of special additional trials for defining the anatomical reference frames [16–18], consistent patterns of joint rotations were rarely observed. These were observed in a previous work by the present authors [21], but the technique involved uncomfortable marker clusters and time-consuming anatomical landmark calibration. The analysis was limited also by the probable rigid motion of the clusters with respect to the underlying bones, and restricted to the first ray of the foot. Many clinical and biomechanical foot concepts, such as foot segment alignments and deviations, medial longitudinal arch angle, navicular drop, first ray mobility, etc., are implied in clinical or radiographic examinations [32,33], but rarely addressed in overall multi-segment foot function analyses in vivo [34–38].

In the present proposal, the selection of the foot segments, the design of the marker set and anatomical reference, and the calculation of the kinematic variables were based on the following clinical interests and general technical indications. In most foot-related functional abnormalities, frontal plane alignment of the rear-foot is essential, both in relation to the shank and the fore-foot [5]. Transverse and sagittal plane deformations of the metatarsus under load have been subject to limited analysis. Deformation under load of the medial longitudinal arch is mostly assessed statically on radiograms or footprints [39–41]. As for the technical design, single markers directly mounted on the skin surface over relevant anatomical landmarks were pursued. More dorsal locations for the fore-foot markers were sought, because of clearance in severely deformed gait, along with locations which can avoid the course of the main foot tendons. In addition, joint lines were used as marker locations, to represent characteristic landmarks of the two adjacent bones, and allow easy and repeatable identification. Other relevant landmarks were calibrated either using an instrumented pointer [42] or an additional marker to be removed before gait trials. The smallest possible marker size was tested, compatible with the additional objective of complete foot marker tracking with the usual number and configuration of TV cameras for full body gait analysis. Finally, clinically oriented definitions of the foot joint rotations were required.

The objective of this work was to design a technique for the in vivo description of ankle and foot joint motion using optoelectronic stereophotogrammetry to be applied in patients with foot pathologies for clinically oriented functional evaluation according to the criteria and goals stated above.

2. Material and methods

2.1. The assumed rigid segments

The following five segments were tracked and assumed to be rigid: (a) the shank which includes tibia and fibula, (b) the foot overall, including all bones, (c) the calcaneus, (d) the mid-foot which includes the navicular, lateral, middle and medial cuneiforms, and the cuboid and (e) the metatarsus which includes the five metatarsal bones (Fig. 1). The proximal phalanx of the hallux, the first, second and fifth metatarsal bones were taken as independent line segments.

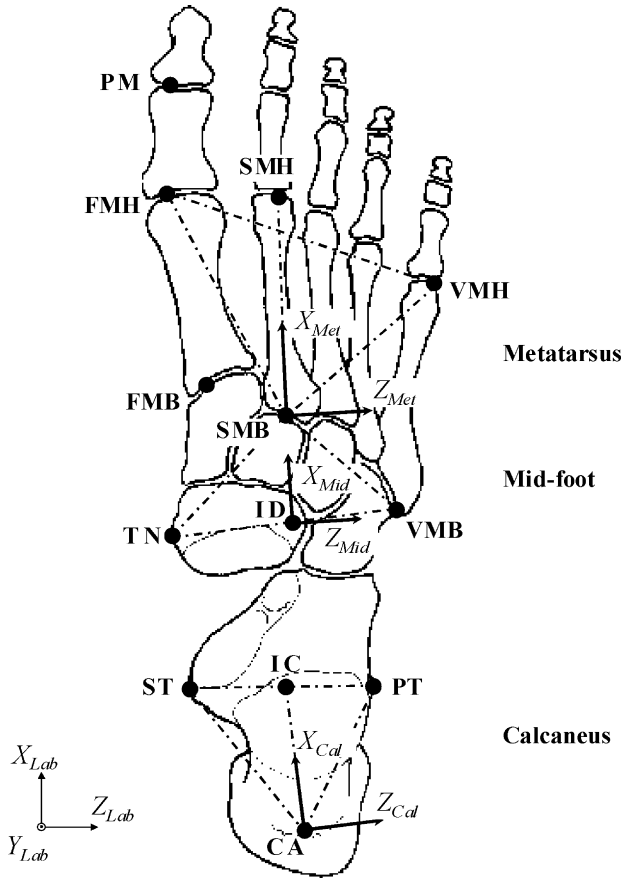


Fig. 1. Diagram of the foot with relevant assumed rigid segments, anatomical landmarks and frames depicted. Transverse planes (dash–dot triangles) and *X*- and *Z*-axes (solid arrows) on these planes are shown, corresponding *Y*-axes pointing proximally.

2.2. The anatomical landmarks

The following anatomical landmarks were tracked in the global photogrammetric frame (Fig. 1). Nearly all of these were identified with direct marker placement, and a few by a simple mid-point calculation (ID, IC and IM) or by a calibration procedure (MM) relative to the segment identified with three other markers (HF, TT and LM).

PM most distal and dorsal point of the head of the proximal phalanx of the hallux.

Metatarsus (Met):

FMB base of the first metatarsal, dorso-medial aspect of the first metatarso-cuneiform joint.

FMH head of the first metatarsal, dorso-medial aspect of the first metatarso-phalangeal joint.

SMB base of the second metatarsal, dorso-medial aspect of the second metatarso-cuneiform joint.
SMH head of the second metatarsal, dorso-medial aspect of the second metatarso-phalangeal joint.
VMB base of the fifth metatarsal, dorso-lateral aspect of the fifth metatarso-cuboid joint.
VMH head of the fifth metatarsal, dorso-lateral aspect of the fifth metatarso-phalangeal joint.

Mid-foot (Mid):

TN most medial apex of the tuberosity of the navicular.
MC the most distal and dorsal aspect of the middle cuneiform, assumed to coincide with SMB.
TC most lateral apex of the tuberosity of the cuboid, assumed to coincide with VMB.
ID intermedius mid-foot, mid point between TN and TC.

Calcaneus (Cal):

CA upper central ridge of the calcaneus posterior surface, i.e. Achilles' tendon attachment.
PT lateral apex of the peroneal tubercle.
ST most medial apex of the sustentaculum tali.
IC intermedius calcaneus, mid point between ST and PT.

Shank (Sha):

TT most anterior prominence of the tibial tuberosity.
HF most proximal apex of the head of the fibula.
LM distal apex of the lateral malleolus.
MM distal apex of the medial malleolus.
IM intermedius malleoli, mid point between LM and MM.

2.3. The anatomical reference frames

According to these landmarks, the following anatomical reference frames were defined (Fig. 1). Orientation of the axes is based on a standardization proposal [43], i.e. *X*-axis pointing forward, *Y*-axis pointing upward, i.e. dorsal on the foot, *Z*-axis pointing to the right.

Metatarsus (Met):

O_{met} the origin is at the SMB.
x_{met} the *x*-axis is the projection of the line joining SMB and SMH on the transverse plane passing through the origin and FMH and VMH.
z_{met} the *z*-axis is orthogonal to *x* and lies in this transverse plane.
y_{met} the *y*-axis is orthogonal to the *xz* plane.

Mid-foot (Mid):

O_{mid} the origin is at ID.
x_{mid} the *x*-axis joins the origin with MC.
z_{mid} the *z*-axis lies in the transverse plane defined by the *x*-axis and TN.
y_{mid} the *y*-axis is orthogonal to the *xz* plane.

Calcaneus (Cal):

O_{cal} the origin is at CA.
x_{cal} the *x*-axis joins the origin with IC.
z_{cal} the *z*-axis lies in the transverse plane defined by the *x*-axis and ST.
y_{cal} the *y*-axis is orthogonal to the *xz* plane.

Right and left foot (Foo), as in ref. [42]

O_{foo} the origin is located at CA.
x_{foo} the *x*-axis is the projection of the line joining the origin and SMH on the transverse plane passing through the origin and FMH and VMH.
z_{foo} the *z*-axis is orthogonal to *x* and lies in this transverse plane.
y_{foo} the *y*-axis is orthogonal to the *xz* plane.

Right and left shank (Sha), as in ref. [42]

O_{sha} the origin is located at IM.
y_{sha} the *y*-axis is the projection of the line joining the origin and TT on the frontal plane passing through the origin and LM and HF.
z_{sha} the *z*-axis is orthogonal to *y*-axis and lies in this frontal plane.
x_{sha} the *x*-axis is orthogonal to the *yz* plane.

2.4. 3D rotations of the joints and planar angles of the line segments

Joint rotations were calculated according to ISB recommendations [44,45], i.e. dorsi-/plantar-flexion (Do/Pl) as the rotation about the *z*-axis (medio-lateral) of the proximal segment, abduction/adduction (Abd/Add) about the *y*-axis (vertical), the distal segment, eversion/inversion (Eve/Inv) about the axis orthogonal to the previous two (floating axis). Motion of the foot with respect to the shank (Sha-Foo), i.e. the ankle complex, the calcaneus with respect to the shank (Sha-Cal), the mid-foot with respect to the calcaneus (Cal-Mid), the metatarsus with respect to the mid-foot (Mid-Met), and also the metatarsus with respect to the not-adjacent calcaneus (Cal-Met) were calculated.

With the proposed marker set, the instantaneous two-dimensional orientation of the line segments representing the first, second and fifth metatarsal bones and the proximal phalanx of the hallux were also calculated. In particular, the following planar angles in the specified anatomical plane (Fig. 2) were calculated:

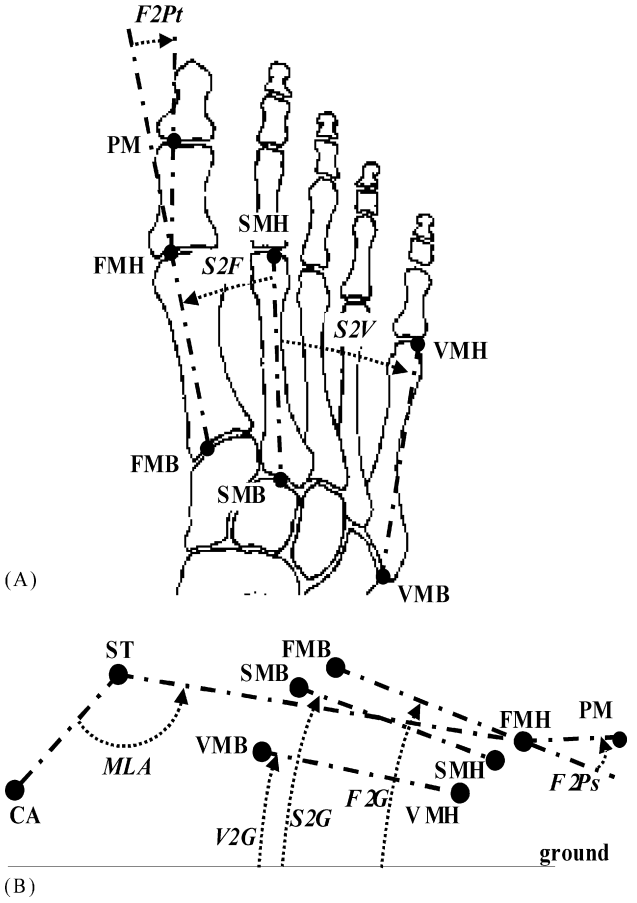


Fig. 2. Scheme for the calculation of the planar angles (dotted arrows, acronyms in italic) in the transverse (A) and sagittal (B) planes. Planar angles are depicted between directions of the hallux and metatarsal line segments (dash–dot). The MLA angle is also depicted between the three relevant landmarks.

F2Pt — angle between the projections of the line segments FMH-PM and FMB-FMH into the transverse plane of the metatarsus, positive angle with the former rotating laterally with respect to the latter, i.e. motion in valgus of the first metatarso-phalangeal joint.

S2F — angle between the projections of the line segments FMB-FMH and SMB-SMH into the transverse plane of the metatarsus, positive angle with the former rotating medially with respect to the latter.

S2V — angle between the projections of the line segments VMB-VMH and SMB-SMH into the transverse plane of the metatarsus, positive angle with the former rotating laterally with respect to the latter.

F2G — angle between the line segment FMB-FMH and the ground, i.e. in the plane orthogonal to the ground and containing FMB-FMH, positive angle with the downward rotation of the head.

S2G — angle between the line segment SMB-SMH and the ground, positive angle with the downward rotation of the head.

V2G — angle between the line segments VMB-VMH and the ground, positive angle with the downward rotation of the head.

F2Ps — angle between the projections of the line segments FMH-PM and FMB-FMH into the sagittal plane of the metatarsus, positive angle with the former rotating upward, i.e. dorsiflexion of the first metatarso-phalangeal joint.

MLA (medial longitudinal arch) — angle between the projections of the line segments CA-ST and ST-FMH into the sagittal plane of the foot, positive angle with clockwise rotation of the former toward the latter, i.e. navicular drop being positive. This would replicate somehow what is known in radiography as the Moreau and Costa Bertani angle [41].

2.5. Experimental procedure

Ten subjects (4 males and 6 females, shoe size in the range 36–43, mean ± standard deviation, age 25.8 ± 4.0

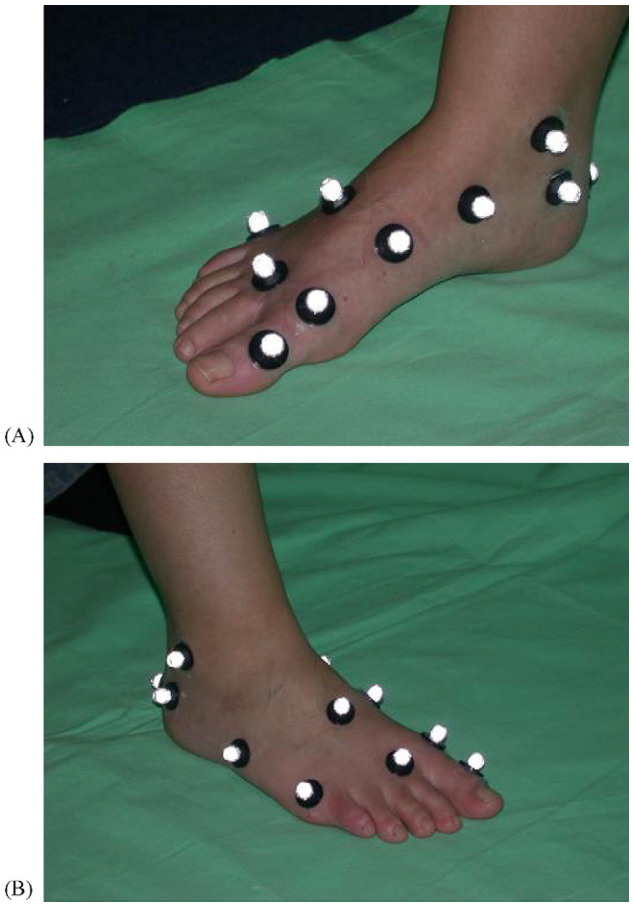


Fig. 3. Antero-medial (A) and antero-lateral (B) views of the marker set at the distal shank and foot.

years, height 167.0 ± 7.6 cm and weight 62.1 ± 11.0 kg), free of any foot deformity or pathology, volunteered for the data collection. The 14 spherical 10 mm diameter skin markers as described in Section 2.2, were mounted over the described anatomical landmarks using double-sided adhesive tape (Fig. 3) by a single examiner with experience in gait analysis. The subjects were first asked to stand up-right in double-leg support and a few samples were collected. This information was used also to calculate offset values for all joint rotations, which were eventually subtracted from the corresponding values over the walking stance. Planar angles were calculated and reported without any subtraction. After checking for a successful acquisition of all marker positions, the MM marker was removed, and the corresponding landmark was reconstructed in space in the walking trials from TT, HF and LM and the local coordinates of MM. The subjects were asked to walk barefoot at their own normal speed, and at least three trials were stored. Marker trajectories and ground reaction force were collected, respectively, by an eight M2-camera motion capture system (Vicon 612, Vicon Motion Systems Ltd., Oxford, UK) and two force plates (Kistler Instrument AG, Switzerland) at 100 samples per second. The camera configuration and calibration procedure was the same as that used for standard full body gait analysis. Ground reaction force was used to identify the stance phase, i.e. the touch-down and the toe-off samples.

3. Results

Joint rotations (Fig. 4) were found to be consistent, and in good agreement with corresponding data obtained with similar anatomical definitions [21], despite the different marker set utilised. It appeared that Sha-Foo kinematics typical of standard gait analysis (first row) was a simplified aggregation of substantial individual joint contributions (second to fourth rows). Considerable motion occurred also out of the sagittal plane (second and third columns).

3.1. 3D rotations of the joints

3.1.1. Shank-foot

Motion in the sagittal plane was very consistent, from a mean of 5° Pl at the loading response phase to a mean of 12° Do at the terminal stance. On the frontal plane, foot was in Inv at initial contact followed by Eve under weight bearing and by Inv again in terminal stance. In the transverse plane, the foot tended to a few degrees Abd until terminal stance, where the foot tended to Add (pre-swing phase).

3.1.2. Shank-calcaneus

In the sagittal and frontal planes, Sha-Cal motion compared well with that of Sha-Foo, apart from the smaller range of Do/Pl which accounted for the corresponding motion at mid-foot and fore-foot. The calcaneus was slightly in Inv at initial contact, then everted under weight bearing until terminal stance when it inverted.

3.1.3. Calcaneus-mid-foot

A small range of motion occurred at the three planes, though very consistent, particularly at the end of stance when the mid-foot plantarflexed, inverted and adducted with respect to the calcaneus.

3.1.4. Mid-foot-metatarsal

As expected, small ranges of motion were observed in the sagittal and transverse planes. The mean 10° Eve at the end of stance resulted from the lateral-to-medial transition of the fore-foot contact to the ground, which was necessary for stabilizing the first metatarsal and for push off.

3.1.5. Calcaneus-metatarsus

The windlass mechanism between rear-foot and fore-foot was detected in the frontal plane. An initial Inv was observed, followed by realignment in mid stance, and by Eve at push off, when the rear-foot was blocked by the constrained mechanism of the subtalar joint. In the sagittal plane, the 15° Pl in late stance was the contribution of the rear-to-fore-foot to achieving the necessary large Do at the metatarso-phalangeal joints at toe-off.

3.2. Planar angles

3.2.1. Transverse plane (Fig. 5)

F2Pt: this angle was at about 5° valgus throughout most of the stance phase. The varus trend at the end might have accounted also for the artefactual motion associated to the large Do at the first metatarso-phalangeal joint.

S2F: this angle remained at about 6° divergence for about half stance, whereas additional 4–5° occurred at terminal stance.

S2V: this angle started from negative values because the fifth metatarsal converged slightly with respect to the second. The angle became less negative and in some subjects even positive, as soon as the fore-foot opened up distally under load in the propulsive phase.

3.2.2. Sagittal plane (Fig. 6)

F2G, S2G, V2G: these showed very consistent patterns in all subjects. The three metatarsals presented an upward orientation in the first 20% of stance, until the full contact of the foot. Interestingly, the inclination of the three metatarsal during stance corresponded to a different height of their head with respect to the ground, already identified in the mean static positions (16°, 25° and 1°, respectively, for the first, second and fifth). At initial contact, the fifth metatarsal was the most inclined upward (−20°) followed by the first (−3.5°) and lastly the second (5°). At about 70% of stance, the metatarsals inverted their orientation with respect to the ground and a mean maximum downward rotation of the head of

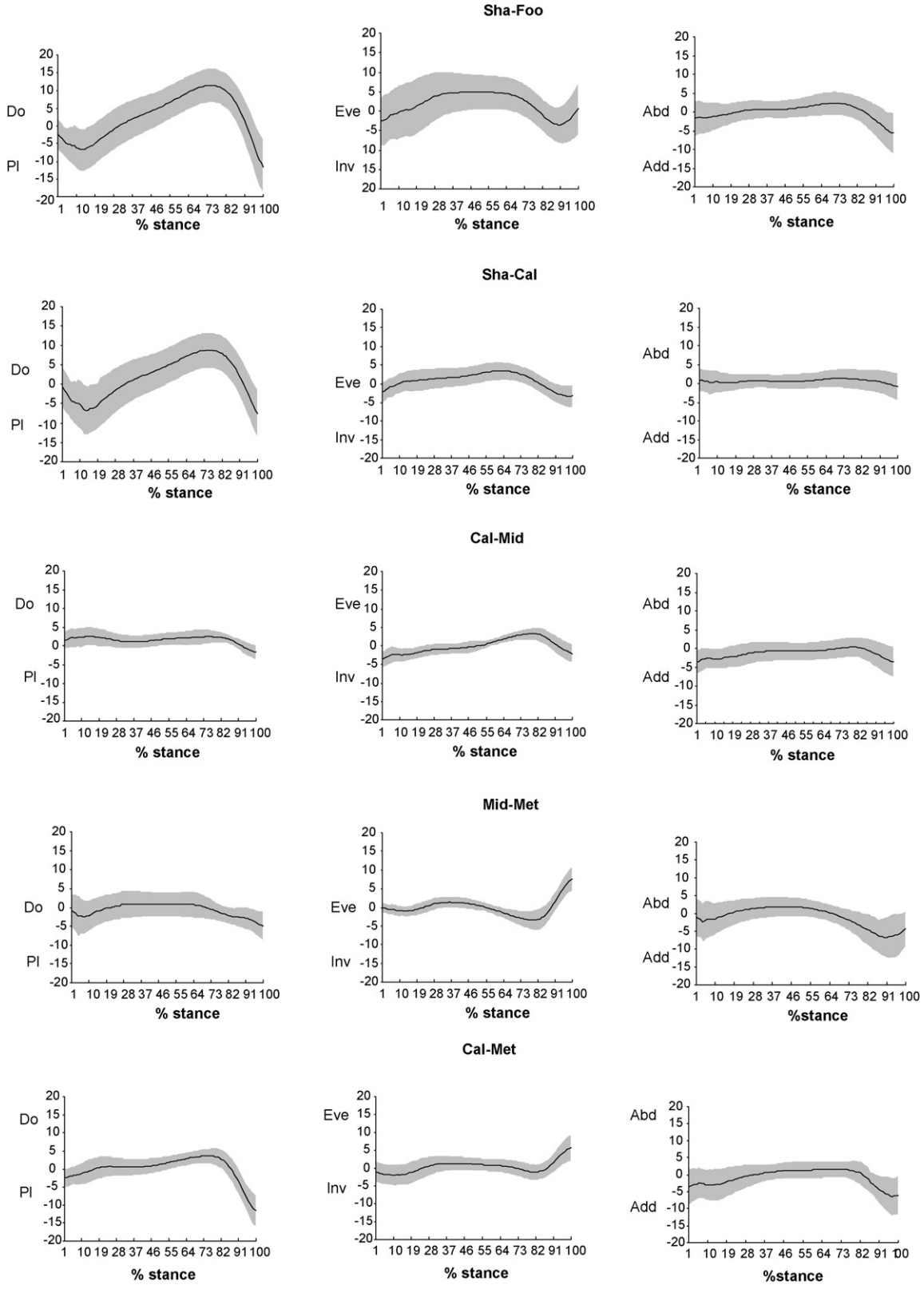


Fig. 4. Rotations about the three joint axes, over the three columns, for the five joints (Sha-Foo, Sha-Cal, Cal-Mid, Mid-Met and Cal-Met), over the five rows. These are reported as mean (solid) plus and minus a standard deviation (grey) over 3 repetitions for all 10 subjects.

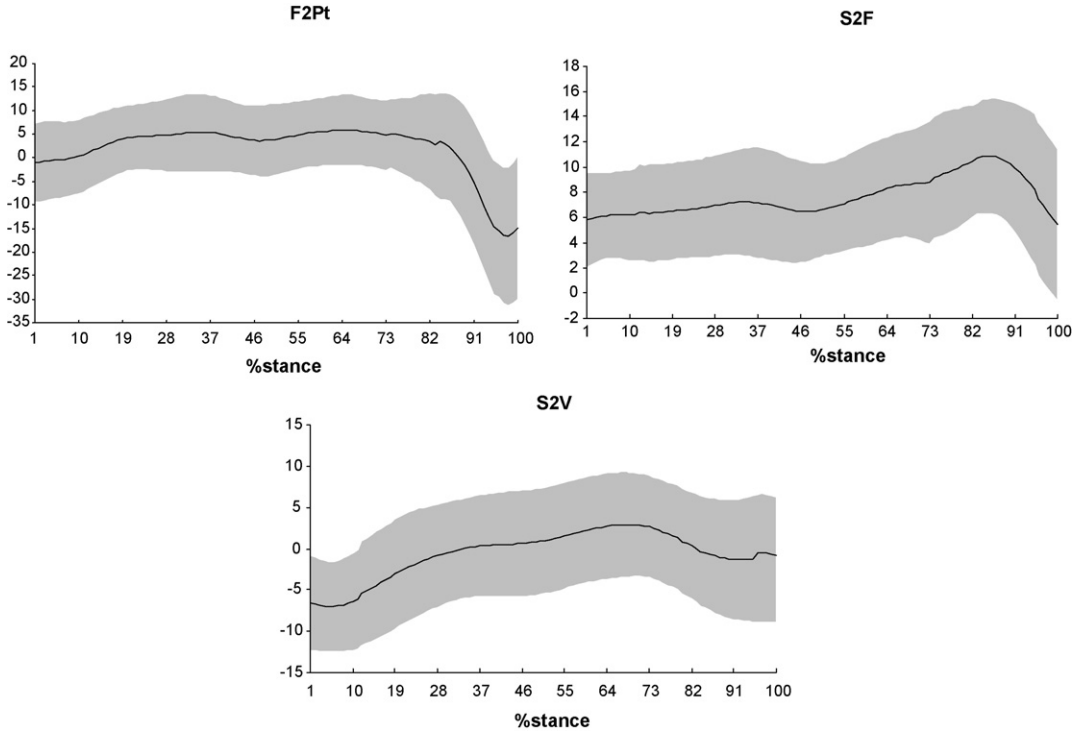


Fig. 5. Planar angles in the transverse plane of the metatarsus, reported as mean (solid) plus and minus a standard deviation (grey) over 3 repetitions for all 10 subjects.

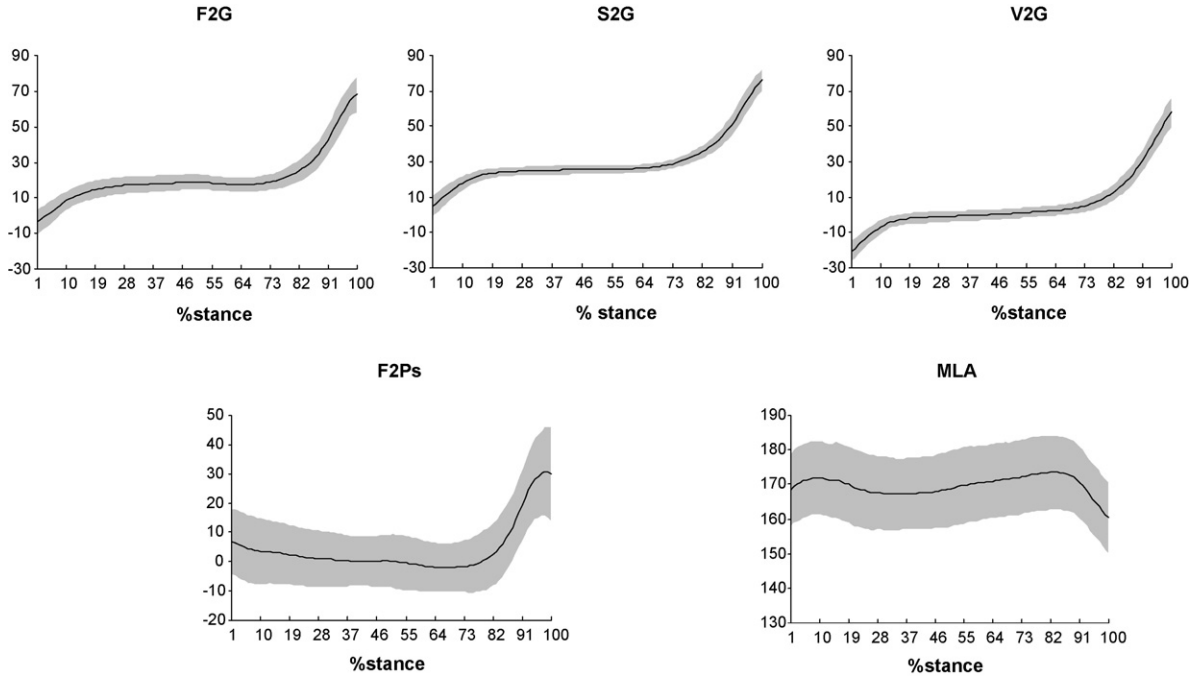


Fig. 6. Other planar angles (see text for acronym explanation), reported as mean (solid) plus and minus a standard deviation (grey) over 3 repetitions for all 10 subjects.

58° for the fifth, 76° for the second and 68° for the first was achieved.

F2Ps: the angle showed slight Do at initial contact (about 7°) which was reduced as soon as the hallux base made contact with the ground. At the end of stance phase, Do increased to about 35° to complete the push off.

MLA: this angle showed a double-bump pattern with an initial 168° followed by a rapid increase under load and then by a reduction in the angle in the single support phase. In the terminal stance, there was a further slight increase and then an evident reduction due to the action of plantar muscles and fascia which tended the arch for the push off.

4. Discussion

The performance of the current motion analysis systems enables the design of advanced protocols for foot segment kinematics in vivo, which are necessary to overcome the single rigid foot segment assumption. Although a very cautious analysis of this motion data is necessary [11,46], the clinical and biomechanical information gained is essential [11]. In particular, the present protocol, in addition to the multi-segment kinematics, was aimed also at measuring the dynamic pattern of angles usually assessed in standard static radiographs, both in the sagittal and transverse planes.

Location of the anatomical landmarks at the fore-foot at relevant joint lines, not only reduced the number of necessary markers, but also facilitated marker mounting. These landmarks do not represent joint centres, but rather articulating points in between two adjacent bones. For the definition of the transverse plane of the metatarsus, six landmarks were available and a number of definitions were possible. Because of the clinical interest in Eve/Inv motion and foot torsion, a single central point proximally and two extreme points distally were chosen to form a triangle converging proximally. In representing sagittal plane motion of the three tracked metatarsals, separate definition of the corresponding projecting planes was preferred to a single reference plane because the latter results in different projection angles just because of the different Abd/Add orientation.

The anatomical landmarks were easily identifiable under the skin by manual palpation, the marker set was fully visible in normal feet, and also in preliminary trials of deformed feet. This marker set was successfully tracked with standard full body TV camera configuration, thus saving the examiners from special camera arrangements and system calibration procedures [26,28,47]. In addition, not only is the marker set proposed here compatible with our original protocol for lower limb kinematics [48], but also the ability of the stereophotogrammetric system to track successfully both relevant marker sets was demonstrated in a few trial tests. Innovatively, the proposed marker set allows the calculation also of the instantaneous two-dimensional orientation of three metatarsals and the hallux. Planar angles suffer from the projection-related artefact [30], particularly over large angles, which can occur in severely deformed feet. However, the metatarsals are close enough in direction to the transverse plane of the metatarsus to limit this computational artefact, and elevation/drop calculation here is irrespective of any other anatomical plane projection. On the other hand, these artefacts may play a considerable role at the first metatarso-phalangeal joint, where about 35° dorsiflexion and 5–7° valgus rotation occurred during stance. However, it is not realistic with this or many other foot marker sets to measure reliably hallux rotation about its longitudinal axis. Finally, the mean Do/Pl and Eve/Inv of this joint in the static posture over the 10 subjects was only $0.2 \pm 5.9°$ and $8.1 \pm 5.9°$, therefore the initial misalignment observed was not severe.

Human motion analysis in vivo is affected by recognized sources of experimental error, i.e. the stereophotogrammetric-based marker position tracking and the skin motion artefact. Although the latter exceedes the former by far, it has been reported that skin motion is much smaller at the foot than at the shank and thigh [46]. Despite the large production of multi-segment foot models, few validation studies have been published. These have been restricted basically to stereophotogrammetric system accuracy [50] and test–retest and inter-rater reproducibility [30]. Although it has been reported that variations on the marker set produces minimal changes in repeatability of the kinematics measurements [28], tests are in progress by the present authors to quantify the sensitivity of calculated rotations to the present marker locations, and to determine the amount of unexpected motion in isolated elementary exercises. As a form of validation, the present results were compared with those of a recent thorough in vitro study which reported on the considerable 3D motion of 15 bones and 22 joints of the foot [49]. In particular, average motion of joints within the mid-foot, here assumed as a single rigid segment, had a maximum of 14.3°, 8.3° and 4.5°, respectively, in the sagittal, frontal and transverse planes. However, the range of motion between the mid-foot and metatarsus compares well with that reported here. Although the present protocol tracks reasonably well the first metatarso-phalangeal joint in the sagittal and transverse planes (respectively $44.2 \pm 8.0°$ and $16.6 \pm 6.1°$ in ref. [49]), the important motion in the frontal plane ($15.4 \pm 7.6°$) is compromised by technical limitations, despite previous attempts using clamps and tripods [19,21,23,27].

The amount of motion identified at the foot joints was considerable, the relevant patterns of motion were very consistent within the single subject, reasonably consistent over the population of 10 volunteers, in good agreement with previous published data, and finally compared well with common clinical observations and established biomechanical knowledge. Furthermore, the values of foot segment rotation of the joints and planar angles of the line segments in the static up-right posture demonstrated a fairly good repeatability in all subjects. In addition to these common measurements, several others which are fundamental for foot function assessment in vivo were taken for the first time in dynamic conditions. The MLA angle is a routine measurement on lateral foot radiograms [41], tracked here throughout the stance phase. It represents the dynamic opening/closing pattern of the medial longitudinal arch, essential for understanding the role of the plantar fascia and the overall flexion ability of the foot joints. This would provide quantitative assessment of relevant pathologies, such as flat or cavus foot. The metatarsal elevation/drop patterns during gait provide new insight into the kinematics of the fore-foot, particularly with respect to the pattern of foot–floor contact and detachment during the stance phase,

typically referred to as the lateral–medial foot loading and unloading patterns and of interest in pathologies such as hammer toes, metatarsalgia and hallux limitus. On the transverse plane the magnitude of the angles between the metatarsals provide useful information in pathologies such as splay-foot. Finally, the concept of the foot as a windlass mechanism can be assessed by means of the proposed motion between rear-foot and fore-foot. In the frontal plane, the pattern observed here compares well with the expected movement in the normal foot.

Despite the fairly large number of models and marker sets described in the literature, there are in fact only a few clinical applications reported in the literature [27,47,51,52]. The present study proposes a possible novel protocol and reports patterns of motion in normal feet, to be exploited soon in the clinical context, as recently recommended [6]. This would hopefully contribute towards a more in depth understanding combining information on foot structure, usually derived from static radiographs, and foot function, examined dynamically by movement analysis. This is fundamental in order to distinguish between normal and pathological foot function, to discriminate between the various levels of impairment, and to assess quantitatively clinical outcomes [53].

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